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The Optimal Global Path Planning of Mobile Robot Based on Improved Hybrid Adaptive Genetic Algorithm in Different Tasks and Complex Road Environments

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ABSTRACT In complex environments, mobile robots performing tasks with different hazard levels need to consider different road factors, this paper proposes a functional model correlating task hazard levels with road factors, proposing an innovative Hybrid Adaptive Genetic Algorithm (HAGA). The HAGA integrates an optimized two-optimization (2-opt) operator* with an enhanced Adaptive Genetic Algorithm (AGA) for efficient path planning in diverse tasks and complex road conditions. Firstly, pre-optimize the initial paths is performed by introducing a new domain knowledge-based operator that duplicates paths in the path are deleted to avoid the redundant paths, and then they are divided into the TOP layer and the ordinary layer, the TOP layer is optimized by using the adaptive 2-opt* operator that satisfies the hyperbolic tangent function (TANH), and the crossover and variability of the ordinary layer are optimized by using the S-type function (Sigmoid function, Sigmoid) and TANH AGA for the optimization treatment for the crossover and variance of the ordinary layer, respectively, to establish a robot path planning algorithm suitable for multitasking in complex environments. The experiment proves that the improved HAGA has strong global search ability and also improves the local search ability, has good generality and robustness, and reduces the optimal path distance by a minimum of 2.74% and a maximum of 10.86% compared with the comparison algorithm in the experiment. The experimental results showed that the method enables mobile robots to perform tasks with different hazard levels in complex environments with good generalization and robustness, and the coefficient of variation (CV) of repeated experiments at five safety levels was kept within 2 %. which had good universality and versatility. This research has the potential to enhance the safety and efficiency of mobile robot operations in challenging environments.

INDEX TERMS Mobile robot, optimal global path planning, hybrid adaptive genetic algorithm, adaptive 2-opt* operator, different tasks optimization model.

I. INTRODUCTION

With the continuous development of science and technology, mobile robots are becoming more important in people's lives and work, such as sweeping robots [1, 2], unmanned aerial vehicles [3], and surface navigators [4]. The prerequisite for mobile robots to perform all kinds of tasks is that they need to have autonomous navigation capabilities, that is, using their own sensors to sense the working environment, obstacles, and their own state position and other information, to establish the

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corresponding environment model, and to plan an optimal path to reach the target point. The navigation technology of mobile robots includes environment perception technology, autonomous positioning and map construction, path planning algorithms, power, and motion control [5].

Path planning [6] is one of the key technologies for mobile robots and is an optimization problem with complexity, nonlinearity, and multi-objective constraints. Path planning algorithms have always been the hotspot of researchers' attention, and after a long period of research and development, different types of path planning algorithms have been continuously proposed and improved, such as the A* algorithm [7], the artificial potential field method [8], the neural network algorithm [9], the genetic algorithm (GA) [10, 11], the RRT algorithm [12], the particle swarm optimization algorithm [13], the ant colony algorithm (ACO) [14, 15], the whale algorithm [16, 17], the grey wolf algorithm [18-20], and the golden jackal optimization algorithm [21]. However, most of the algorithm optimization objectives do not consider enough factors, easily fall into local minima, and have slow convergence.

The Genetic algorithm is a global optimization method based on natural heredity and natural selection, which is widely used to solve path planning problems due to its strong adaptability to different spatial environments and high robustness, but the algorithm's local search ability is poor, and the search efficiency is low in the late evolutionary stage. Nazarahari M [22] proposed the use of augmented genetic algorithms for multi-objective multi-robot path planning in continuous environments. Tuncer A [23] proposed a new mutation operator in GA for premature convergence. Tsai CC [24] proposed a parallel elite GA for smoothing mobile robot paths. Hu CA [25] proposed a hybrid GA that is robust to external perturbations. Lee J [26] proposed an efficient population initialization method for GAs based on directed acyclic graphs. Although the method has a wide range of applications, the quality of the initial population obtained is poor when the coverage of obstacles in the environment is low. In addition, the method employs a greedy method in the generation of the directed acyclic graph, which reduces the diversity of the population, thus potentially leading to the premature convergence of the algorithm.

The current mobile robot for path safety is less considered and stays away from obstacles to a greater extent; only a few scholars consider the complex road situation. Li XW [27] proposed the consideration of road potholes, road surface attachment coefficient, and other road conditions under the function of the state of the road. Ji XY [28] for the problem of road undulation, presented a combination of the improvement of the A* algorithm and the dynamic window of the optimization algorithm. The complexity of the road is crucial for the safety of traveling mobile robots and is a key factor of path planning. For complex roads, not only the intricacy of the road surface should be considered, such as pothole condition, road surface attachment coefficient, height and height difference of the road surface, but also the classification of the road, such as lawn road, sandy road, concrete road, and asphalt road, and other factors. Therefore, different road conditions correspond to diverse safety levels, and the paths planned under different safety levels are also dissimilar. The current path planning is for a specific kind of task situation, so only the corresponding kind of road safety needs to be considered. However, for different tasks, mobile robots need to consider different path safety situations. According to the diverse hazard levels of the task, the need to consider the safety level of the mobile robot traveling path is also different. Mobile robots also need to reach their destinations quickly and efficiently considering safety issues.

To address the above problems, this paper proposes a functional model correlating task hazard levels with road factors, proposing an innovative Hybrid Adaptive Genetic Algorithm (HAGA). The main contributions of this paper are: 1) Pre-optimize the initial paths is performed by introducing a new domain knowledge-based operator that duplicates paths in the path are deleted to avoid the redundant paths. 2) The path population is divided into a TOP layer and an ordinary layer. The TOP layer is optimized using an adaptive two-opt* (2-opt*) operator under TANH, which is mainly aimed at increasing the ability of the local search, and the crossover and variation of the ordinary layer are optimized using an adaptive 2-opt* operator under Sigmoid and AGAs under TANH with the main purpose of increasing the convergence speed of the algorithm. 3) A new function model is proposed to link the tasks of different risk levels with road conditions. Mainly through the complex traffic function model to set different weight values, so as to achieve the task risk level and traffic factors linked.

The rest of the paper is structured as follows: Section 2 is the model creation and corresponding mathematical description of the map environment required for the experiments. Section 3 is the algorithm section, which focuses on the improvement work of the GA innovations and the establishment of the multi-objective function model. Section 4 is the experimental section, which focuses on experimental demonstration, comparison of the methodologies and data analysis, and discussion of the results. Section 5 is the concluding section, which summarizes the work accomplished and the organization of and outlook on the next step of the work.

II. ENVIRONMENTAL MODELING AND MATHEMATICAL DESCRIPTION

The robot path planning problem is to search for an optimal path that satisfies the conditions under specific constraints, which can be given as

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$$\min f(p), \forall p \subseteq \Re(p_s, p_o) \tag{1}$$

In Equation (1), *f(p)* denotes the objective function of the optimized path, p is the positional path, p_s is the start position, p_o is the end position, and $\Re$ (p_s, p_o) denotes the set of all feasible path nodes between the start position and the end position.

Environment modeling refers to the mapping of the actual environment onto an abstract space, and map construction is an important part of path planning. The main methods of map environment construction are the raster map method, the visual map method, the free space method, the topological map method, and the semantic map method. Because raster maps are easy to create and manage, and each location corresponds to a unique point, they are used to model the mobile robot environment. Rasterizing the map is easy. Through the raster for the description of the map elements, the map is equally divided into a number of small rasters to ensure the safety of the robot's operation. The actual environment is converted into a raster map process stipulating obstacles occupying less than a raster will be expanded into a complete raster, as shown in Figure 1 In the raster map, the black area is the obstacle raster, representing the obstacles, and the white area is the free raster, which can be selected as a path point.

The raster map environment is shown in Figure 2 In the raster map environment, the raster number is increased in order from bottom to top and from left to right. For example, Figure 2 shows the raster number in the lower left corner is 0, which indicates the starting position of the path, and the raster number in the upper right corner is 399, which indicates the ending position of the path.

The correspondence between the nth raster and the coordinates (x_n, y_n) of its center is given as

$$\begin{cases} x_n = \mathrm{mod}(n, M) + 1 \\ y_n = \mathrm{fix}(n / M) + 1 \end{cases} \tag{2}$$

$$n = (x_n - 1) + (y_n - 1) * M \tag{3}$$

In Equations (2) and (3), $n = 0, 1, \cdots, N$; $N = 399$, $M = 20$ in Figure 2; (x_n, y_n) is the center coordinate of the first raster; M is the number of rows of the raster, mod is the remainder operation; and fix is the rounding operation to zero. Equation (2) is used to convert the grid number into grid coordinates, and Equation (3) is used to transform the grid coordinates into grid numbers.

The raster distance is calculated, the current raster is defined as (x_i, y_i), the next raster is (x_{i+1}, y_{i+1}), and its position is judged, which can be given as

$$\begin{cases} |x_i - x_{i+1}| + |y_i - y_{i+1}| = 1 \\ |x_i - x_{i+1}| + |y_i - y_{i+1}| \neq 1 \end{cases} \tag{4}$$

In Equation (4), if the neighboring grid calculation result is 1, the two adjacent grids are up and down, left and right for the four cases; if the neighboring grid calculation result is not 1, the two adjacent grids have a diagonal relationship (Example: diagonal direction of the proximity grid), and the distance is $\sqrt{2}$.

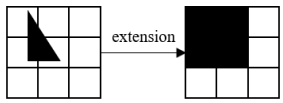


FIGURE 1. Principle of raster map extension.

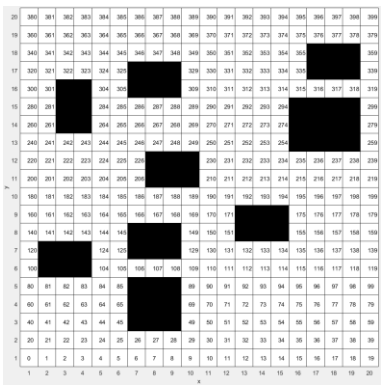


FIGURE 2. 20×20 raster map.

III. IMPROVEMENT AND MODELING OF GENETIC ALGORITHMS

In the 1970s, John Holland [29, 30] in the United States proposed the GA, which is designed based on the laws of biological evolution in nature, and its core idea is to simulate the natural evolutionary process to search for the optimal solution. When solving with a GA, the set of all practical solutions to the problem is called the population, and each solution in the population is called an individual (also known as a chromosome).

The general process of the traditional GA consists of operations such as generating the initialized number of populations N, calculating the individual fitness function values, selection, crossover, and mutation, as shown in Figure 3.

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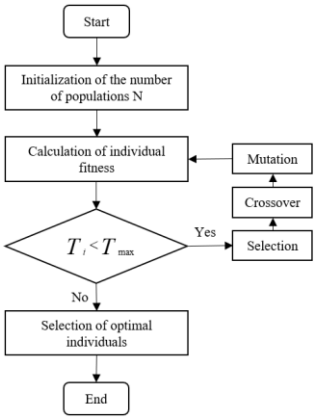


FIGURE 3. Flowchart of genetic algorithm.

A. INITIALIZING POPULATIONS

Population initialization refers to initializing a feasible population to meet the requirements of optimization before the evolutionary optimization process. The purpose is to make the population more characteristic of the population itself and to be able to meet the expectation values of all feasible solutions in the space of all fitness functions. In the optimization process, the initialization of the population can effectively help the search of the optimization algorithm to obtain the optimal solution.

The population initialization of the genetic algorithm refers to the initial gene expression assigned to the individuals in the population before the algorithm begins to execute. Its purpose is to ensure the diversity of individuals as much as possible so that a wider search space can be searched in the subsequent evolutionary process.

The GA starts the iteration with an initialized population as the initial point, the size of the initial population indicates the number of individuals contained in the population, each individual represents a chromosome, and the length of the chromosome consists of the number of rasters in each path. During population initialization, we randomly select one or more free grids inside the optional free grids in each row to generate a continuous unobstructed path, thus forming individuals in a population, and multiple individuals form a population. Initialization requires randomly selecting one free raster as a path point in each row of rasters in sequence, except for the rows where the start and end points are located, and then connecting the start point, path point, and end point to form a continuous unobstructed path, as shown in Figures 4 and 5, respectively. The main steps to initialize the path are as follows:

Step 1: Determine the rows where the start and end points are located;

Step 2: Randomly select a free raster as a path point in each row other than the start and end rows;

Step 3: Determining whether the path points in two adjacent rows are adjacent points;

Step 4: Perform path point supplementation between path points that are not neighboring points, and make them into continuous obstacle-free paths by adding the free grids of the neighboring points.

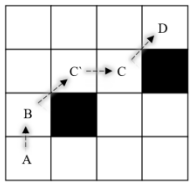


FIGURE 4. Initialization path.

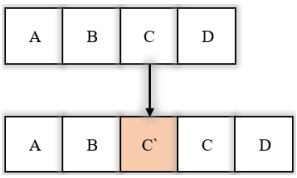


FIGURE 5. Generating a continuous unobstructed path.

B. PRE-OPTIMIZE PROCESSING

Because the initialization population is a continuous unobstructed path formed by randomly generated path points in sequence, this inevitably leads to the possibility of having identical path points in a path. Identical path points increase not only the length of the path but also its computation, in order to reduce the amount of computation, improve the optimization efficiency and speed up the convergence, we pre-optimize the initialized population. Based on this problem, we propose a new operator based on domain knowledge, which mainly uses the existing population information to achieve optimization of the initialized population by discriminating the path points of each path through path point information search. By introducing new operators based on domain knowledge, the initial paths are pre-optimized to remove duplicate paths in the paths and avoid redundant paths, as shown in Figure 6. The main steps of the pre-optimization process are as follows:

Step 1: For each path in the order of the path points, determine whether the same point exists at each point in turn;

Step 2: Find all the identical points if they exist;

Step 3: Delete operation between the first point and the last point in the identical points;

Step 4: Continue to determine the next point, repeat steps 1 to 3, until all the path points are judged.

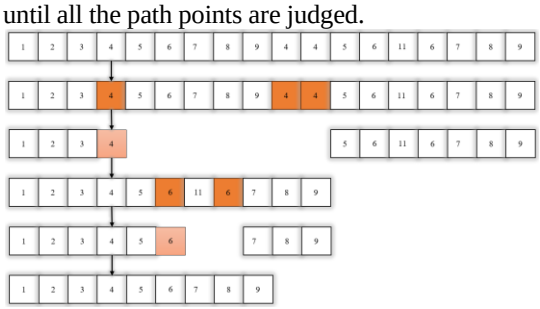


FIGURE 6. Pre-optimized processing.

C. IMPROVING THE FITNESS FUNCTION

The fitness function in the GA represents the quality of individuals in the population, which responds to the

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probability that an individual can be selected for the next generation, and indicates the fitness status of an individual under the constraints, which is used to clarify the evolutionary direction of the population. In the traditional GA, the fitness function is only related to the path distance, which can be given as

$$l = \sum_{i=1}^{n-1} \sqrt{(x_{i+1} - x_i)^2 + (y_{i+1} - y_i)^2} \tag{5}$$

Equation (5) only considers the length of the path, which cannot adapt to the complex outdoor road surface, to improve the comprehensive performance of the path; according to the multi-objective fitness function, the path can be optimized from multiple aspects [31]. For the planning of the path to reduce the energy consumption as much as possible, the safety factors under different road conditions of diverse road surfaces as well as the effect of the danger of different tasks on the path selection must be considered to optimize the path under the influence of these comprehensive factors. Finally, a different tasks optimization function model under complex outdoor road conditions is established as the fitness function, which can be given as

$$F(k) = \omega_1 \frac{1}{L(k)} + \omega_2 \frac{1}{T(k)} + \omega_3 \frac{1}{H(k)} + \omega_4 \frac{1}{S(k)} \tag{6}$$

In Equation (6), *L(k)* is the path distance function of the *k* path, *T(k)* is the turning energy consumption function of the *k* path, *H(k)* is the pavement height function of the *k* path, *S(k)* is the complex road condition function of the *k* path, and ω_1, ω_2, ω_3, ω_4 are the weights corresponding to each of the four indicators.

1) PATH DISTANCE FUNCTION

For the path planning problem, the foremost concern is the distance problem. Finding the optimal path among many paths is the first problem to consider, and the shortest path that satisfies the other constraints must be found among all the eligible paths, which will be used in the fitness function can be given as

$$L = \min \sum_{i=1}^{n-1} \sqrt{(x_{i+1} - x_i)^2 + (y_{i+1} - y_i)^2} \tag{7}$$

In Equation (7), *n* is the number of all rasters through which the path passes, (x_i, y_i) is the current raster position, and (x_{i+1}, y_{i+1}) is the position of the next raster.

2) TURNING ENERGY CONSUMPTION FUNCTION

In addition to the requirement of the shortest distance of the path, the mobile robot considers the energy consumption due to turning. The excessive number of turns leads to the robot's energy loss. Similarly, the size of the arc of the turn affects the loss of energy, and the larger the arc of the turn is, the more energy consumed. Energy consumption and cost are positively correlated, so the energy loss caused by turning needs to be considered in path optimization. By adding a penalty function, the number of turns is reduced and the energy loss is reduced. Therefore, the turn energy consumption function is added to the fitness function, which can be given as

$$\chi_i = (x_{i+2} - x_i)^2 + (y_{i+2} - y_i)^2, i \in [1, n-2] \tag{8}$$

$$t_i = \begin{cases} 5, 4 < \chi_i < 8 \\ 50, 1 < \chi_i < 4 \\ 5000, \chi_i \le 1 \\ 1, else \end{cases} \tag{9}$$

$$T = \sum_{i=1}^{n-2} t_i \tag{10}$$

In Equations (8)-(10), (x_i, y_i) is the current grid position, (x_{i+2}, y_{i+2}) is the position of the last two grids, χ_i is the square of the straight line distance between the current position and the last two positions, t_i is the penalty value of the current position, and *T* is the penalty value of all the current path. When $4 < \chi_i < 8$, the rows (columns) of the two grids are adjacent to one another, the columns (rows) are one grid apart, and the turning arc is acute. When $1 < \chi_i < 4$, the rows and columns of the two grids are adjacent to one another, and the turning arc is right angle. When $\chi_i \le 1$, the rows (columns) of the two grids are adjacent to one another, or they are the same position, and the turning arc is greater than 90° ; the other means that it is traveling in a straight line, no turning is performed, and no turning energy loss is observed. Therefore, the penalty function value is the minimum.

3) PAVEMENT HEIGHT FUNCTION

In the actual operating environment of a mobile robot, the surrounding road surface is not always flat, and changes in height are often encountered. The road surface environment height change is often caused by the road surface inclination, that is, the emergence of slope. The slope can be classified according to size as flat, slightly sloping, low sloping, medium sloping, and high sloping. The change of slope often brings certain dangers and is prone to cause the robot to roll over and shake during operation, affecting the safety of robot operation. Therefore, when encountering environments with substantial changes in gradient, the number of climbing times should be minimized or climbing should be avoided directly to reduce or avoid the dangers brought about by changes in gradient. To make the path smoother and safer, the pavement height function is added to the adaptation function, which can be given as

$$h_i = \begin{cases} 0, \text{flat land} \\ 1 \sim 4, \text{ slight slope} \\ 6 \sim 8, \text{ low slope} \\ 10 \sim 12, \text{ medium slope} \\ \ge 13, \text{ high gradient} \end{cases} \tag{11}$$

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$$\kappa_i = \frac{h_{\max} - (h_{i+1} - h_i)}{h_{\max} - h_{\min} + 0.01}\lambda + \delta \tag{12}$$

$$H = \sum_{i=1}^{n-1} \kappa_i \tag{13}$$

In Equations (11)-(13), h_i is the height value and slope classification of the current raster, h_{i+1} is the height value and slope classification of the next raster, h_{max} is the maximum height difference between the current raster and the surrounding rasters, h_{min} is the minimum height difference between the current raster and the surrounding rasters, λ, δ is the height correction constant, 0.01 ensures that the denominator is not 0, and κ_i is the combined height impact value of the current raster.

4) COMPLEX ROAD FUNCTION

In the actual robot operating environment, the road is often not a fixed surface but may be grass, sand, gravel, concrete, or asphalt. Similarly, the road conditions are not fixed. The road conditions in real environments are complex and varied, and water or potholes may exist, resulting in waterlogged potholes, waterlogged smooth pavement, dry potholes, wet smooth pavement, and dry smooth pavement. The gravel road surface causes the robot to jitter when running. The lawn road surface and the sandy road surface cause the robot to slip to different degrees during operation. The waterlogged and wet road surface reduces the adhesion between the robot and the ground to different degrees, reducing the friction, which in turn causes the robot to slip. Potholes in the road surface also cause the robot to jitter when running, which affects the robot's operational safety and service life. Different road conditions have dissimilar effects on the safety of the robot. Gravel, potholes, or waterlogged roads should be avoided to improve the safety of robot operation. Thus, the complex road condition function is added to the fitness function, which can be given as

$$\varphi_i = \begin{cases} 0, \text{ asphalt road} \\ 1, \text{ concrete road} \\ 5, \text{ grass road} \\ 10, \text{ sandy road} \\ 15, \text{ gravel road} \end{cases} \tag{14}$$

$$\phi_i = \begin{cases} 0, \text{ dry and smooth road surface} \\ 1, \text{ Wet and smooth road surface} \\ 5, \text{ ponding and smooth road surface} \\ 10, \text{ dry and pothole road surface} \\ 15, \text{ ponding and pothole road surface} \end{cases} \tag{15}$$

$$S = \sum_{i=1}^{n} (\alpha_1 \varphi_i + \alpha_2 \phi_i + \beta) \tag{16}$$

In Equations (14)-(16), φ_i is the calculated value of the current raster with the pavement classification, ϕ_i is the calculated value of the current raster with the pavement road condition, α_1, α_2 is the weight value corresponding to the two indicators, respectively, and β is the road condition correction constant.

D. MULTITASKING FUNCTION

In the actual operation of mobile robots, the surrounding environment is complex and variable, the magnitude of the influence of different environments on the safety of robot traveling is different, and the environmental effect is positively correlated with the hazardous situation of the task. Due to the complexity of the environment, as well as the diversity of tasks, we allow mobile robots to have a wider range of operating scenarios and be able to operate safely and efficiently in complex and changing environments by associating the hazards of different tasks with complex environments. The tasks of the robot are diverse, such as transporting chemical reagents, military ammunition, fragile items, ordinary courier, fruit, and grain. The robot may also perform different task contents, such as sweeping, delivery, field military, outdoor inspection, transporting different things, and performing different tasks, and the need to consider the road safety level is also different.

The method of judging the danger of an item [32] is based on a combination of four aspects: 1) the impact on human health, 2) the impact on the environment, 3) its own chemical properties, 4) other hazards. Based on the categorization of the items, thus the hazard level is classified for the situation of the task performed by the robot.

According to the situation of the task, the danger level of the task is classified into five levels: no-dangerous tasks, such as sweeping the floor; low-dangerous tasks, such as delivery of food and courier; general-dangerous tasks, such as outdoor inspection; higher-dangerous tasks, such as the transport of fragile items; and high-dangerous tasks, such as transporting chemical reagents and military munitions, which can be given as

$$Task = \begin{cases} \text{I, Non-hazardous missions} \\ \text{II, Low-risk missions} \\ \text{III, General dangerous missions} \\ \text{IV, higher risk missions} \\ \text{V, High-risk missions} \end{cases} \tag{17}$$

For the task safety levels assigned in Equation (17), the variations of the corresponding road traveling heights and the traveling complex road parameters of the road surface are shown in Table 1.

TABLE 1. Driving under the mission security level.

Task	Traveling height	Traveling on complex roads	ω_3, ω_4 variations	λ, δ, β variations
I	$h_i \le 16$	$\varphi_i \le 15$ & $\phi_i \le 15$	ω_3, ω_4 small	λ, δ, β large

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II	$h_i \le 12$	$\varphi_i \le 10$ & $\phi_i \le 10$	ω_3, ω_4 comparativ ely small	λ, δ, β comparativ ely large
III	$h_i \le 8$	$\varphi_i \le 5$ & $\phi_i \le 5$	ω_3, ω_4 moderate	λ, δ, β moderate
IV	$h_i \le 4$	$\varphi_i \le 1$ & $\phi_i \le 1$	ω_3, ω_4 comparativ ely large	λ, δ, β comparativ ely small
V	$h_i = 0$	$\varphi_i = 0$ & $\phi_i = 0$	ω_3, ω_4 large	λ, δ, β small

E. HIERARCHICAL OPTIMIZATION PROCESSING

When traditional genetic algorithms carry out selection operations, the excellent individuals are lost or destroyed by mutation operations, in order to avoid the loss of the optimal solution in the process of population iteration, we adopt a layered operation, dividing the population into the TOP layer and the ordinary layer, which can increase the diversity of the population while retaining the excellent individuals, and effectively prevent it from falling into the local optimal solution. The TOP layer improves the excellent individuals individually and retains them to the next generation, ensuring that the advantageous genes can be inherited to the next generation as much as possible. The general layer uses traditional genetic operations, mainly to generate new individuals, increase the diversity of the population, and prevent the algorithm from falling into local optimal solutions.

Hierarchical operation optimization is to calculate the size of fitness of each individual, then sort the individuals in descending order of fitness from the largest to the smallest, and then divide the population into two parts, the former part is the TOP layer, and the latter part is the ordinary layer.

In order to prevent the TOP layer from falling into a local optimum, we use a local search method for it to increase the likelihood of producing better offspring. The TOP layer is optimized using the adaptive 2-opt* operator [33] under TANH [32]. 2-opt* is an inter-path improvement method, which belongs to the local search algorithm, which takes a point randomly among the same points of two paths and exchanges the path points from that point to the end point with each other. The exchange method is shown in Figure 7.

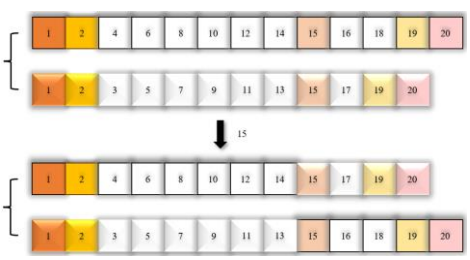


FIGURE 7. 2-opt* operator exchange method.

For the optimization operation of the 2-opt* operator, its local search probability remaining unchanged leads to its inability to adapt to the complex environment. To improve the search performance and efficiency of the algorithm, adaptive local search probabilities are used in the local search. The general adaptive approach has two main methods. One is to change with the number of iterations adaptively, and the other is to change according to the size of individual fitness adaptively. At present, most scholars adopt the second adaptive method but ignore the influence of the number of iterations; if only the first adaptive method is used, it will be limited by the number of iterations. Based on this problem, this paper adopts TANH to combine the characteristics of the two approaches. Firstly, a temporary probability is obtained through the individual fitness size calculation in Equation (18), and then this temporary probability is calculated through the TANH Equation (19) to determine the final local search probability.

$$pn_int = \begin{cases} pn_l, f_m = f_a \\ \dfrac{pn_h * (f_m - f)}{f_m - f_avg}, f \ge a, and, f_m \ne f_a \\ pn_h, f < f_a, and, f_m \ne f_a \end{cases} \tag{18}$$

$$pn = pn_int * \frac{e^{t/T} - e^{-t/T}}{e^{t/T} + e^{-t/T}} \tag{19}$$

In Equations (18) and (19) *pn_l* is the minimum local search probability, *pn_h* is the maximum local search probability, *f_m* is the maximum fitness of the population, *f_a* is the average fitness of the population, *f* is the current individual fitness, *pn_int* is the temporary local search probability, *pn* is the final local search probability, *t* is the current number of evolutionary generations, and *T* is the total number of evolutionary generations.

F. GENETIC SELECTION

Selection operation refers to the selection of good individuals from an old population with a certain probability to form a new population, the purpose of selection is to make the population continuously retain the good individuals and maintain the uniform diversity of paths, which in turn makes the population continuously evolve. The probability of an individual being selected is related to the size of the fitness, the greater the fitness of an individual, the greater the probability of being selected. In order to realize the survival law of the survival of the fittest and let the excellent individuals have a greater probability to be retained in the evolutionary process, in this paper, the roulette method is used for random selection operation, and the probability of an individual being selected is positively correlated with the size of the fitness value, which can be given as

$$p(x_i) = \frac{f(x_i)}{\sum_{j=1}^{N} f(x_j)} \tag{20}$$

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$$q(x_i) = \sum_{j=1}^{i} p(x_j) \tag{21}$$

Equation (20) calculates the probability of each individual being selected, and Equation (21) accumulates the probabilities for each individual. It can be represented by the probability wheel in Figure 8.

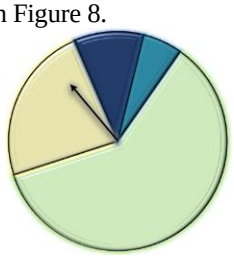


FIGURE 8. Roulette wheel of probability.

G. ADAPTIVE CROSSOVER

Crossover operation is an important operation in GA, where genes are recombined through crossover, thus increasing the diversity of the population and improving the global search ability to evolve the population. Crossover is the process of randomly selecting two individuals from a population and passing on the superior characteristics of the previous generation to the next generation by exchanging combinations of the two chromosomes, thus producing new superior individuals. To avoid destroying the good genes of the previous generation in the crossover recombination, this paper adopts the two-point crossover method, which improves the randomness of the algorithm and prevents the algorithm from falling into the local optimal solution while guaranteeing the good genes are not destroyed. The operation is as follows: Firstly, whether the two path individuals have at least two identical points except the starting point, that is, at least three identical points is determined. If they exist, then two identical points are randomly chosen among the identical points except the starting point to exchange to produce a new next-generation individual; if three or more identical points do not exist, the two paths are not exchanged and are directly preserved for the next-generation individual. The crossover is shown in Figure 9.

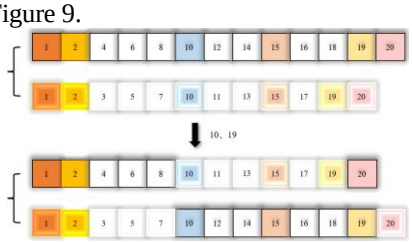


FIGURE 9. Cross operation.

The choice of crossover probability has a crucial effect on the efficiency of the algorithm's operation and the speed of convergence. In the basic GA, the crossover probability is fixed. If the crossover probability is larger, the individuals in the population can evolve at a faster rate, but the structure of the individuals with higher adaptation will be destroyed. If the crossover probability is smaller, the individuals in the population will evolve slowly. The research of other scholars finds the population is extraordinarily rich in the early stage, so a larger crossover probability is needed but later stabilized. The crossover probability needs to be reduced, and a smaller crossover probability must be used. Excellent individuals need to be protected, their crossover reduced, and a smaller crossover probability set. For the others, a larger crossover probability can speed up their evolution. To improve the search performance and efficiency of the algorithm, we design an adaptive crossover probability. In order to prevent the crossover probability equal to 0, we use Sigmoid to combine the characteristics of the two methods. First, a temporary probability is derived through the individual fitness size calculation Equation (22), and then the final crossover probability is computed by calculating this temporary probability through the Sigmoid Equation (23).

$$pc_int = \begin{cases} pc_l, f_m = f_a \\ \dfrac{pc_h * (f_m - f)}{f_m - f_a}, f \geq f_a, and, f_m \neq f_a \\ pc_h, f < f_a, and, f_m \neq f_a \end{cases} \tag{22}$$

$$pc = pc_int * \frac{1}{1 + e^{-t/T}} \tag{23}$$

In Equations (22) and (23), *pc_l* is the minimum crossover probability, *pc_h* is the maximum crossover probability, *f_m* is the maximum population fitness, *f_a* is the average population fitness, *f* is the current individual fitness, *pc_int* is the interim crossover probability, *pc* is the final crossover probability.

H. ADAPTIVE MUTATION

Mutation operation is a key part of GA, where genes mutate one or part of the chromosomes in an individual through mutation to produce new individuals. In mutation, to produce more good individuals. In this paper, we adopt a two-point mutation method, which can enhance the mutation effect of the algorithm, effectively improve the evolutionary speed and local search ability of the population, and maintain the diversity of the population, relative to the traditional single-point mutation method. The operation is as follows: First, any two points are randomly selected in the path individuals except the start point and the end point, the path between the two points is deleted, and then the path points are randomly generated to complement it as a continuous unobstructed path. The mutation is shown in Figure 10.

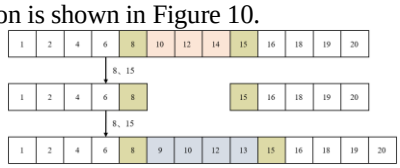


FIGURE 10. Mutation operation.

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The choice of mutation probability has an important effect on the local search ability of GAs. In the basic GA, the probability of mutation is fixed. If the probability of mutation is too low, it will affect the algorithm's ability to search locally. If the probability of mutation is too high, it will make the evolution of the population too random, which is not conducive to the retention of the best individuals. Similar to crossover probability, a small mutation probability is used in this paper because the population is extraordinarily rich in the early stages, but a large mutation probability needs to be used because the population richness decreases in the later stages. Similarly, the best individuals need to be protected by adopting a smaller mutation probability, and a larger mutation probability must be adopted for others to speed up their evolution. In order to improve the search performance and efficiency of the algorithm and to prevent the GA from falling into a local optimum, we design an adaptive mutation probability. Based on this, TANH is used to combine the characteristics of the two approaches by first obtaining an interim probability through the individual fitness size Equation (24) and then calculating this interim probability through the TANH Equation (25) to obtain the final crossover probability.

$$pm_int = \begin{cases} pm_l, f_m = f_a \\ \dfrac{pm_h * (f_m - f)}{f_m - f_a}, f \geq f_a, and, f_m \neq f_a \\ pm_h, f < f_a, and, f_m \neq f_a \end{cases} \tag{24}$$

$$pm = pm_int * \frac{e^{t/T} - e^{-t/T}}{e^{t/T} + e^{-t/T}} \tag{25}$$

In Equations (24) and (25), *pm_l* is the minimum probability of variation, *pm_h* is the maximum probability of variation, *f_m* is the maximum fitness of the population, *f_a* is the average fitness of the population, *f* is the current individual fitness, *pc_int* is the interim probability of variation, *pc* is the final probability of variation.

I. IMPROVED GENETIC ALGORITHM USED IN THIS PAPER

By improving and innovating the traditional GA, a new GA, namely, improved HAGA, is finally obtained as shown in Figure 11.

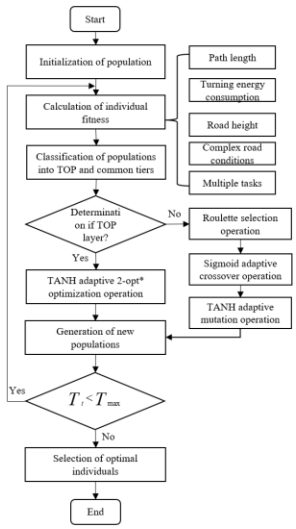


FIGURE 11. Improved hybrid adaptive genetic algorithm.

IV. VALIDATION OF EXPERIMENTAL METHODS

To validate the improved HAGA for path planning method of different task robots under complex roads and to demonstrate the performance and robustness of each algorithm better, this paper conducts comparative experiments in 3D directions. The first dimension is the comparison experiment on the performance of the algorithms. The algorithms in this paper are compared with other different algorithms, and the comparison experiments are conducted through the raster maps in different dimensions and under different numbers of obstacles in the same dimension to judge whether the improved algorithms are superior to the previous methods to reflect the wide application, generality and robustness of the algorithms as a whole. The second dimension is the comparison experiment in different environmental factors, in which the algorithm of this paper is compared with the algorithm without considering environmental factors, only considering the factor of road height, the factor of complex road conditions, and the factor of road height and complex road conditions to compare whether the algorithm is effective. The third dimension is the comparison experiment under different tasks, in which the algorithm of this paper compares the operation of the tasks under five safety levels considering road safety to verify whether the algorithm is dependable under different task levels. To make the experimental results more objective and fairer, and to avoid chance phenomenon, all the experiments in this paper are repeated no less than 20 times to make the results more convincing and to better reflect the broadness of the conclusions. The setting of the parameters in this paper mainly refers to the design experience of other scholars [27], as well as observing whether the experimental results converge or not, mainly through the standard deviation of the path length of 20 repetitions of the experiment, the standard error and IQR and other aspects of the data results to comprehensively judge the reasonableness of the setting of the parameters in this paper.

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A. EXPERIMENTAL ENVIRONMENT

To complete the various comparative experiments mentioned above, the experimental environment used in this paper is shown in Table 2.

TABLE 2. The hardware and software configuration.

Hardware	Processor	12th Gen Intel(R) Core (TM) i7-12700H 2.30 GHz
	RAM	16 GB (15.7 GB available)
Software	Operating system	Windows 11 (64-bit operating system)
	Simulation tool	MATLAB R2022a

B. ALGORITHM PERFORMANCE ANALYSIS

In order to verify the superiority of the HAGA used in this paper in terms of algorithmic performance, and in order to reflect the wide range of applications, generality and robustness of the algorithm as a whole. The following comparison is illustrated in three main aspects. The first aspect is to compare the different algorithms for comparison experiments. Traditional GAs, AGA, ACOs, and algorithms (HAGA) used in this paper are analyzed. The second aspect is a comparison experiment under different numbers of obstacles in the same grid dimension analyzed under 10–20, 30–40, and 70–80 obstacles. The third aspect is to conduct comparison experiments under different raster dimension sizes of 10×10, 30×30, and 40×40 rasters. The results of the experiments are analyzed 20 times.

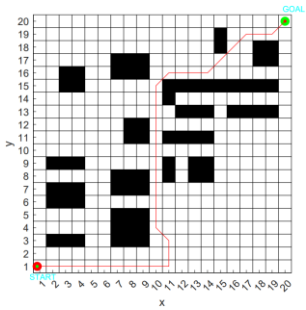
1) DIFFERENT ALGORITHMS

This subsection is a comparative analysis of different algorithms: traditional GA, AGA, ACO, and improved HAGA. The experimental environment is a 20×20 and 74 obstacles. The experimental parameters are shown in Table 3, and the path planning results of different algorithms in the raster map are shown in Figure 12.

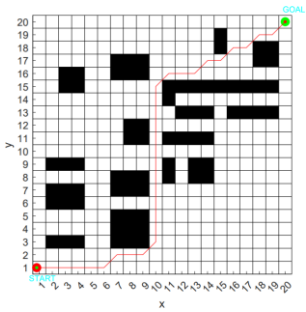
Alpha is the importance degree of pheromone, *Beta* is the importance degree of heuristic factors, and *Rho* is the evaporation coefficient of pheromone.

TABLE 3. Parameter settings for different algorithms.

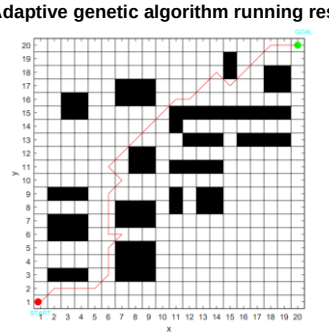
Parameter	GA	AGA	ACO	HAGA
N	200	10	200	200
T	100	100	100	100
pc	0.8			
pc_max		0.8		1
pc_min		0.6		0.8
pm	0.2			
pm_max		0.25		0.8
pm_min		0.1		0.6
pn_max				0.9
pn_min				0.8
Alpha			1	
Beta			7	
Rho			0.3	



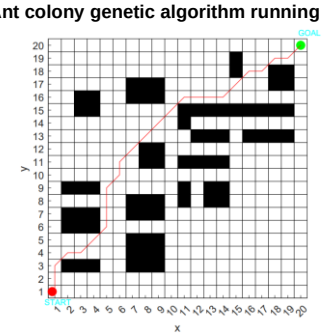
(a). Traditional genetic algorithm running results.



(b). Adaptive genetic algorithm running results.



(c). Ant colony genetic algorithm running results.



(d). Improved hybrid adaptive genetic algorithm operation results.

FIGURE 12. Results of running different algorithms.

Figure 12 shows the improved HAGA used in this paper is substantially better than the three other methods in terms of finding the optimal path, and the previous three algorithms have a certain number of duplicate paths and backtracking phenomenon in the selection of local paths, which in turn causes the global optimal path to be longer than the optimal path found by this paper's algorithm, which indicates that this paper's algorithm has a better ability to search for the optimal path both locally and globally. This shows that the algorithm in this paper has better local and global optimisation ability. At the same time, we repeated the experiment no less than 20 times to get the distance value of the optimal path each time,

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and took out the 20 consecutive distance values for data analysis, the distance unit (m), to get the results of the experimental data in Table 4.

TABLE 4. Running data of different algorithms.

Algorithm Names	Mean ± Standard Deviation	Variance	Median	Standard Error	IQR	Coefficient of Variation(CV)
GA	35.579±3.176	10.088	35.314	0.710	5.328	0.710%
AGA	32.715±2.594	6.730	31.849	0.580	3.368	0.580%
ACO	32.607±1.216	1.478	32.627	0.272	1.538	0.272%
HAGA	31.714±0.624	0.389	31.678	0.139	1.172	0.139%

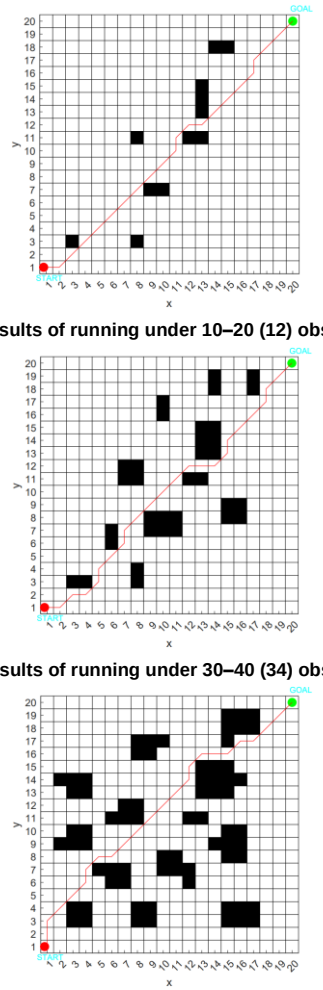
In Table 4, analyzing the mean, standard deviation, variance, median, standard error, IQR, and CV of the 20 path planning results in a multi-dimensional manner reveals the various data of the HAGA path planning results used in this paper are remarkably better than those of the three other algorithms. It is proved that the algorithm has better global optimization ability and stability. The minimum distance of the optimal path is reduced by 2.74 %, and the maximum is reduced by 10.86 %. At the same time, the CV after repeated experiments was also the smallest, at 0.139 %.

2) DIFFERENT NUMBERS OF OBSTACLES

This subsection is a comparative analysis of different numbers of obstacles; the obstacles compared are 10–20, 30–40, and 70–80; the experimental environment is 20×20 raster; and the experimental algorithm is HAGA. The experimental parameter settings are shown in Table 5, and the algorithm's path planning results for raster maps with different numbers of obstacles are shown in Figure 13.

TABLE 5. Parameter settings for different numbers of obstacles.

Parameter	HAGA
N	200
T	100
pc_max	1
pc_min	0.8
pm_max	0.8
pm_min	0.6
pn_max	0.9
pn_min	0.8



(a). Results of running under 10–20 (12) obstacles.

(b). Results of running under 30–40 (34) obstacles.

(c). Results of running under 70–80 (71) obstacles.

FIGURE 13. Running results with different numbers of obstacles.

Figure 13 shows the improved HAGA used in this paper can perform a better optimization search under different numbers of obstacles. Through no less than 20 repeated experiments, the distance value of each optimal path is obtained, and 20 consecutive distance values are taken out for data analysis. The distance unit (m) is obtained, and the experimental data results in Table 6 are obtained.

TABLE 6. Operational data for different numbers of obstacles.

Number of Obstacles	Mean ± Standard Deviation	Variance	Median	Standard Error	IQR	CV
10~20	28.042±0.000	0.000	28.042	0.000	0.000	0.000%
30~40	29.257±0.544	0.296	29.213	0.122	0.243	1.860%
70~80	30.060±1.082	1.171	29.506	0.242	1.414	3.600%

In Table 6, multidimensional analysis of the data from 20 runs with different numbers of obstacles in the same raster dimension finds that as the number of obstacles increases, the paths grow with them, while the standard deviation, variance,

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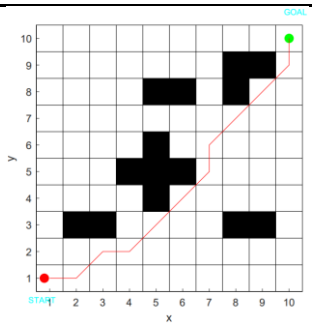
standard error, IQR and CV of the path remain in a reasonable range of changes, and the overall change is small, with good stability and universality. Overall, the HAGA used in this paper can find the optimal or better paths under different obstacles.

3) DIFFERENT RASTER DIMENSION SIZES

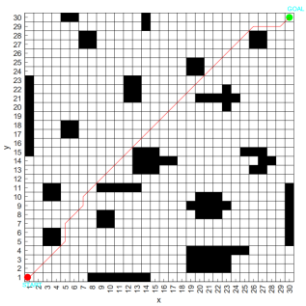
This subsection is a comparative analysis of different raster dimensions. The raster dimensions for comparison are 10×10, 30×30, and 40×40, and the experimental algorithm is HAGA. the experimental parameters are shown in Table 7, and the algorithm's path planning results for raster maps in different raster dimensions are shown in Figure 14.

TABLE 7. Parameter settings for different raster dimension sizes.

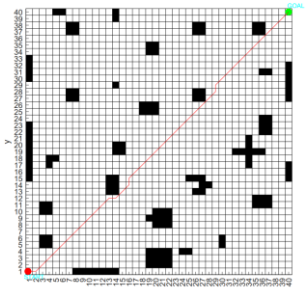
Parameter	10×10	30×30	40×40
N	200	200	200
T	100	200	300
pc_max	1	1	1
pc_min	0.8	0.8	0.8
pm_max	0.8	0.8	0.8
pm_min	0.6	0.6	0.6
pn_max	0.9	0.9	0.9
pn_min	0.8	0.8	0.8
start	0	0	0
goal	99	899	1599



(a). Running results at 10×10 grid.



(b). Running results at 30×30 grid.



(c). Running results at 40×40 grid.

FIGURE 14. Running results with different raster dimensions.

Figure 14 shows the improved HAGA used in this paper can perform satisfactory optimization search under different grid dimensions. Through no less than 20 repeated experiments, the distance value of each optimal path is obtained, and 20 consecutive distance values are taken out for data analysis. The distance unit (m) is obtained, and the experimental data results in Table 8 are obtained.

TABLE 8. Running data in different raster dimensions.

Raster Dimension	Mean ± Standard Deviation	Variance	Median	Standard Error	IQR	CV
10×10	14.034±0.499	0.249	13.899	0.112	0.586	3.554 %
30×30	43.443±0.567	0.321	43.355	0.127	0.586	1.305 %
40×40	56.384±0.262	0.069	56.326	0.059	0.000	0.465 %

In Table 8, analyzing the data of 20 runs of the raster map path planning results in different dimensions in a multi-dimensional manner finds that as the raster dimension increases, the paths grow with them, while the standard deviation, variance, standard error, IQR and CV of the path remain in a reasonable range of changes, and the overall change is small, with good stability and universality. Overall, the HAGA used in this paper can find optimal or better paths under different raster dimension sizes.

C. ANALYSIS OF ENVIRONMENTAL FACTORS

To verify whether the algorithm of this paper is general in different environmental factors, the following comparison is illustrated in four main cases. The first one is an experiment without considering the environmental factors, the second one is an experiment in the case of considering only the pavement height, the third one is an experiment in the case of considering only the factors of complex road conditions (including the pavement classification and the condition of the pavement), and the fourth one is an experiment under the influence of the combined factors of considering the pavement height and the complex road conditions.

The experimental parameters are shown in Table 9, and the experiment analyzes the results of 20 sessions. The experimental environment is a 20×20 grid and 40 obstacles. The path planning results of different environmental factors in the raster map are shown in Figure 15.

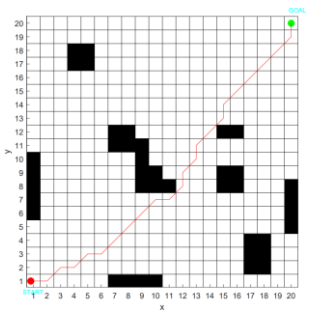
TABLE 9. Parameter settings in different environments.

Parameter	No consideration of the environment	Road height	Complex road conditions	Overall factors
N	200	200	200	200
T	100	100	100	100
pc_max	1	1	1	1
pc_min	0.8	0.8	0.8	0.8

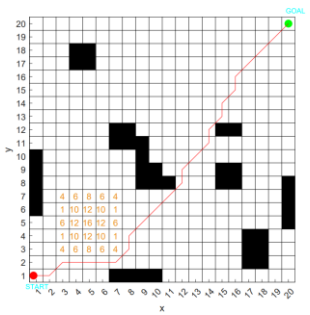
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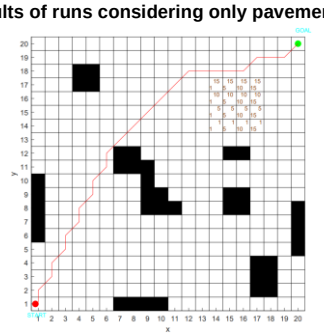
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pm_min	0.6	0.6	0.6	0.6
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pn_min	0.8	0.8	0.8	0.8
ω_1	8	8	8	8
ω_2	2	2	2	2
ω_3	0	8	0	8
ω_4	0	0	10	15
λ	0	3	0	3
δ	0	1	0	1
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α_2	0	0	0.5	0.5
β	0	0	1	1



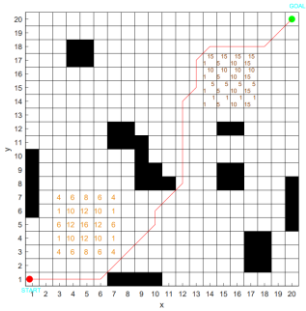
(a). Operational results without environmental considerations.



(b). Results of runs considering only pavement height.



(c). Operational results considering only complex road conditions.



(d). Operational results considering pavement height and complex road conditions.

FIGURE 15. Results of operation under different environmental factors.

Figure 15 shows the HAGA used in this paper can find the optimal path under different environmental factors. At the same time, we obtain the distance value of each optimal path through no less than 20 repeated experiments, and take out the 20 consecutive distance values for data analysis, the distance unit (m), and obtain the experimental data in Table 10.

TABLE 10. Operational data under different environmental factors.

Environmental Factors	Mean ± Standard Deviation	Variance	Median	Standard Error	IQR	CV
No consideration of the environment	29.448±0.481	0.231	29.213	0.108	0.000	1.633%
Road height	29.770±0.299	0.089	29.799	0.067	0.000	1.004%
Complex road conditions	30.669±0.996	0.992	30.385	0.223	0.439	3.248%
Overall factors	33.208±2.930	8.583	32.728	0.655	3.075	8.822%

In Table 10, through the multi-dimensional analysis of the data of 20 path planning results, we can see that the length of the path changes with the increase of the environmental factors considered. The standard deviation, variance, standard error, IQR and CV of the path remain in a reasonable range of changes, and the overall change is small, with good stability and adaptability. Overall, the HAGA used in this paper can find optimal or better safe paths under different environmental factors.

D. ANALYSIS OF THE SITUATION OF THE MANDATE

To verify whether the algorithm in this paper can find the optimal path safely in different task situations, the following experimental analysis is mainly conducted from five task situations. According to the different danger levels of the five tasks, different path safety levels are divided, and diverse optimal paths under dissimilar safety levels are found.

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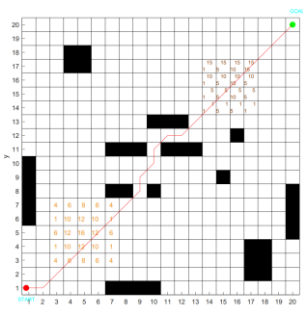
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Therefore, based on other previous experiments, the data recording and analysis of the maximum road surface height, maximum road surface classification calculation value, and maximum road surface road condition calculation value in the driving path are added.

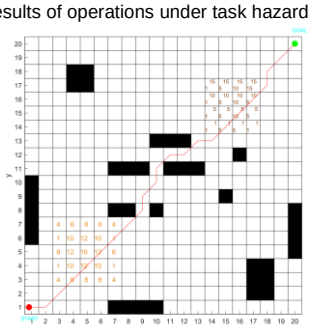
The experimental parameters are shown in Table 11, and the experiment analyses the results of 20 sessions. The experimental environment is a grid of 20×20 and 37 obstacles. The path planning results of different task hazard classes in the raster map are shown in Figure 16.

TABLE 11. Parameter settings for different task hazard classes.

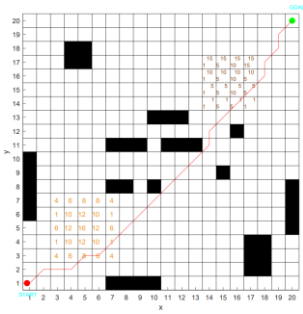
Parameter	Security class I	Security class II	Security class III	Security class IV	Security class V
N	200	200	200	200	200
T	100	100	100	100	100
pc_max	1	1	1	1	1
pc_min	0.8	0.8	0.8	0.8	0.8
pm_max	0.8	0.8	0.8	0.8	0.8
pm_min	0.6	0.6	0.6	0.6	0.6
pn_max	0.9	0.9	0.9	0.9	0.9
pn_min	0.8	0.8	0.8	0.8	0.8
ω_1	6	6	6	6	6
ω_2	2	2	2	2	2
ω_3	2	4	6	8	12
ω_4	2	4	5	8	12
λ	15	12	9	4	1
δ	10	8	6	3	1
α_1	0.5	0.5	0.5	0.5	0.5
α_2	0.5	0.5	0.5	0.5	0.5
β	10	8	6	3	1



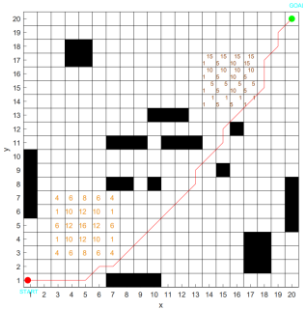
(a). Results of operations under task hazard class I.



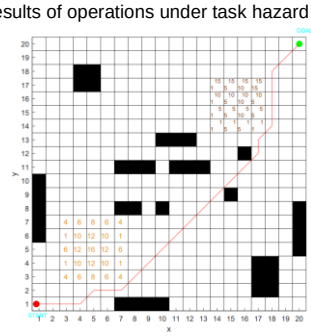
(b). Results of operations under task hazard class II.



(c). Results of operations under task hazard class III.



(d). Results of operations under task hazard class IV.



(e). Results of operations under task hazard class V.

FIGURE 16. Operational results at different task hazard levels.

Figure 16 shows the HAGA used in this paper can find the optimal paths that satisfy the safety conditions under different task danger levels. At the same time, we get the distance value of each optimal path through no less than 20 repeated experiments, and take out the 20 consecutive distance values for data analysis, the distance unit (m), get the experimental data of Table 12 and Table 13.

TABLE 12. Operational data for different mission hazard classes.

Task level	Mean ± Standard Deviation	Variance	Median	Standard Error	IQR	CV
I	28.042±0.000	0.000	28.042	0.000	0.000	0.000%
II	28.686±0.262	0.069	28.627	0.059	0.000	0.913%
III	29.348±0.572	0.327	29.799	0.128	1.172	1.949%
IV	29.916±0.240	0.058	29.799	0.054	0.000	0.804%
V	30.013±0.325	0.105	29.799	0.073	0.586	1.082%

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TABLE 13. Path safety data for different task hazard classes.

Task level	Maximum road height	Calculated maximum pavement classification	Calculated maximum pavement condition
I	h_{man}=12	φ_{max}=15	ϕ_{max}=15
II	h_{man}=12	φ_{max}=10	ϕ_{max}=10
III	h_{man}=8	φ_{max}=5	ϕ_{max}=5
IV	h_{man}=0	φ_{max}=1	ϕ_{max}=1
V	h_{man}=0	φ_{max}=0	ϕ_{max}=0

Tables 12 and 13 show by multidimensional analysis of the data of 20 times path planning results that the length of the path increases with the increase of the task hazard level; the maximum pavement height, the maximum pavement classification computed value, and the maximum pavement road condition computed value decrease; and the corresponding safety performance increases gradually. The standard deviation, variance, standard error, IQR and CV of the path remain in a reasonable range, and the overall change is small. The CV is kept within 2 %, which has good universality and stability. Overall, the HAGA used in this paper can find the optimal or better safe paths that satisfy the corresponding safety level under different task hazard classes.

V. CONCLUSIONS AND OUTLOOK

In order to relate the tasks with different hazard levels to road condition factors, this paper designs an improved HAGA algorithm. By establishing a new complex road condition function model as the adaptation degree of the genetic algorithm, we realize that the mobile robot performs the tasks of different danger levels in complex environments with good generality and robustness. For the problem of insufficient local optimization ability of genetic algorithm, we effectively improve its ability in local optimization by adding an adaptive 2-opt* operator, etc. We compare the improved HAGA with different algorithms, and after comparing the path mean, variance, IQR, CV, etc. for 20 times, we found that the path planning results using the HAGA algorithm are significantly better than other comparison algorithms, and there is no repeated path and back-to-back phenomenon. The optimal path distance is reduced by 2.74% at the minimum and 10.86% at the maximum. At the same time, the CV after repeated experiments is also the smallest, which is 0.139%, which effectively proves that HAGA has better global optimization ability and stability. Through multi-dimensional experiments in different numbers of obstacles, different grid dimension sizes and different environmental factors, it is found that the path planning results of the method used in this paper have smaller variance, IQR, CV, etc., which illustrates that the method has good generalization and robustness. In this paper, by setting different weights for the complex road condition function model, thus realizing the linking of the task danger level with the road condition factor, the research may improve the safety and efficiency of mobile robots operating in complex environments. All the experiments in this paper are based on known static environments, and at the same time, due to the reason of time, we did not carry out a more detailed study on the task hazard class and road condition factors, the next step will be to carry out experiments in dynamic and unknown environments, and at the same time, we will carry out a more in-depth study on the categorization of the task hazard class and road condition factors.

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